Arc length of polar curves — 7.4

For a parametric curve, the length of the curve from t_1 to t_2 is

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt.$$

For a polar curve, $x(\theta) = f(\theta)\cos(\theta)$, $y(\theta) = f(\theta)\sin(\theta)$.

$$L = \int_{\theta_1}^{\theta_2} \sqrt{\left(f'(\theta)\cos(\theta) - f(\theta)\sin(\theta)\right)^2 + \left(f'(\theta)\sin(\theta) + f(\theta)\cos(\theta)\right)^2}\, d\theta$$

$$L = \int_{\theta_1}^{\theta_2} \sqrt{\left(f'(\theta)\cos(\theta) - f(\theta)\sin(\theta)\right)^2 + \left(f'(\theta)\sin(\theta) + f(\theta)\cos(\theta)\right)^2}\, d\theta$$

$$= \int_{\theta_1}^{\theta_2} \sqrt{f'(\theta)^2\cos^2\theta - 2f'(\theta)f(\theta)\cos(\theta)\sin(\theta) + f(\theta)^2\sin^2\theta + f'(\theta)^2\sin^2(\theta) + 2f'(\theta)f(\theta)\sin(\theta)\cos(\theta) + f(\theta)^2\cos^2(\theta)}\, d\theta$$

$$= \int_{\theta_1}^{\theta_2} \sqrt{f'(\theta)^2\left(\cos^2(\theta) + \sin^2(\theta)\right) + f(\theta)^2\left(\sin^2(\theta) + \cos^2(\theta)\right)}\, d\theta$$

$$= \int_{\theta_1}^{\theta_2} \sqrt{f'(\theta)^2 + f(\theta)^2}\, d\theta = \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2}\, d\theta$$

Example: Find the length of the cardioid $r = 1+\cos(\theta)$.

$$L = \int_{\theta_1}^{\theta_2} \sqrt{f'(\theta)^2 + f(\theta)^2}\, d\theta = \int_0^{2\pi} \sqrt{(-\sin(\theta))^2 + (1+\cos(\theta))^2}\, d\theta$$

$$\boxed{\cos^2(x) = \frac{1+\cos(2x)}{2}}$$

$$= \int_0^{2\pi} \sqrt{\sin^2(\theta) + 1 + 2\cos(\theta) + \cos^2(\theta)}\, d\theta$$

$$= \int_0^{2\pi} \sqrt{2+2\cos(\theta)}\, d\theta = \int_0^{2\pi} \sqrt{4\cos^2(\theta/2)}\, d\theta$$

$$= \int_0^{2\pi} 2\left|\cos(\theta/2)\right| d\theta = \int_0^{\pi} 2\cos(\theta/2)\, d\theta - \int_{\pi}^{2\pi} 2\cos(\theta/2)\, d\theta$$

$$2\left(2\sin(\theta/2)\Big|_0^{\pi} - 2\sin(\theta/2)\Big|_{\pi}^{2\pi} \right)$$

$$= 2\left(2\sin(\tfrac{\pi}{2}) - 2\sin(0) - 2\sin(\pi) + 2\sin(\tfrac{\pi}{2})\right) = 2(2-0-0+2) = 8.$$